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Permian–Triassic boundary magnetostratigraphy from the Southern Alps (Italy)

Robert Scholger [a,*], Hermann J. Mauritsch [a], Rainer Brandner [b]

[a] *Institute of Geophysics, University of Leoben, Peter-Tunner-Strasse 25, 8700 Leoben, Austria*
[b] *Institute of Geology and Palaeontology, University of Innsbruck, Innrain 52, 6020 Innsbruck, Austria*

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Abstract

Palaeomagnetic investigations of Permian–Triassic boundary sections in the Dolomites provided a magnetostratigraphy for the uppermost part of the Permian Bellerophon Formation to the Lower Triassic Werfen Formation. Magnetite was the dominant magnetic component in most of the samples, while the presence of hematite was characteristic for the Tesero horizon, which is regarded as the immediate Permian–Triassic boundary layer. The palaeomagnetic results are consistent with earlier studies of Permian and Triassic sediments and volcanites in the Southern Alps. The mean characteristic remanence directions indicate deposition and remanence acquisition at a shallow northerly palaeolatitude and a counterclockwise rotation. A zone of reversed polarity occurs at the top of the Permian sequence and a polarity change from reverse to normal polarity shortly above the base of the Werfen Formation. The onset of a tectonically controlled sedimentary cycle marks the beginning of the basal Triassic normal polarity zone, which corresponds to most of Griesbachian time. The record of geomagnetic polarity across the Permian–Triassic boundary confirms the apparent continuity as found by sedimentologic biostratigraphic investigations which emphasise a gradual lithic transition with no evidence for a time gap at the boundary. © 2000 Published by Elsevier Science B.V. All rights reserved.

Keywords: Permian-Triassic boundary; paleomagnetism; magnetostratigraphy; Dolomites; Tethys; Europe

1. Introduction

The boundary between the Palaeozoic and Mesozoic represents the greatest destruction of the marine ecosystem within the geological record. Except for a few sections in Greenland, Asia and South China, the Permian–Triassic boundary is normally a stratigraphic unconformity around the globe [1]. In the Southern Alps, the Permian–Triassic boundary is located between uppermost Bellerophon Formation and lowermost Werfen Formation, but opinions differ as to the exact chronostratigraphic position and the type of contact at the Bellerophon–Werfen boundary [2]. Sedimentologic investigations emphasise a gradual lithic transition with no evidence for a time gap at the boundary [3–5]. The apparent continuity prompted us to obtain a re-

* Corresponding author. Fax: +43-3126-50414; E-mail: scholger@unileoben.ac.at

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cord of geomagnetic polarity across the Permian–Triassic boundary and the Lower Triassic in sections in the Southern Alps.

Changes of the polarity of the Earth's magnetic field are world wide phenomena, enabling correlation and dating of geological material. Magnetic polarity stratigraphy, the stratigraphic record of polarity reversals in rocks and sediments, is central to the construction of geologic time scales, linking biostratigraphies, isotope stratigraphies and absolute ages [6]. The magnetostratigraphic record of the Late Palaeozoic is characterised by a long period of reverse polarity, the Permo–Carboniferous reversed polarity superchron [7,8], which ends within the Late Permian. The reverse superchron (also called Kiaman reversed superchron) is followed by a period of mixed polarity, the Illawarra mixed polarity superchron, with at least seven normal polarity zones in the Late Permian [9]. This polarity change from reverse to normal polarity near the base of the Triassic is known as the Illawarra reversal [10]. Magnetostratigraphic investigations of sediment successions formed in different environments yielded apparently contradictory results as to the exact time of the onset of the Illawarra reversal, because geomagnetic reversal time-scales are based on absolute age determinations, which were often missing. A record of the Illawarra reversal was recently found within the Upper Permian Tatarian stage in continental red beds on the eastern Russian platform [11]. This result compares with other Late Permian polarity sequences, derived from sections of marine carbonates in Pakistan and Transcaucasia [12].

Magnetostratigraphic studies across the Permian–Triassic boundary indicate a polarity change from reverse to normal polarity at or shortly above the base of the Triassic. According to the sedimentologic investigations in the Southern Alps, a possible record of this reversal in the studied sections in Bulla and Siusi was expected to occur within the Tesero Horizon of the Werfen Formation. Unfortunately, it has become apparent that Permian–Triassic boundary sections often contain important unconformities. However, a thin zone of reversed polarity is observed at the top of Permian sequences in many regions of the world [1,12–18] and the subsequent basal Triassic normal polarity zone corresponds to most of Early Griesbachian time [6]. Environmental changes at the Permian–Triassic boundary are also evident from pronounced signals in stable isotope records which have been observed world-wide [19–22]. In particular, the carbon isotope signal shows a significant drop across the boundary.

2. General geological situation

Research in the Dolomites has produced a series of rather complete Permian–Triassic boundary sections indicating a palaeogeographic position in the inner-shelf area of a basin opening eastwards. The coastline of the Permian Bellerophon basin was located approximately 25 km west of Siusi, while the Triassic Werfen sea reached beyond this area westwards (Fig. 1). The basin development is characterised by transcurrent movements in the Lower and Middle Permian, and Mid-Triassic [23]. Tectonic quiescence and a constant subsidence rate predominate along the Permian–Triassic boundary interval. The Permian–Triassic boundary is located within a depositional sequence bounded by unconformities of Mid-Permian and Mid-Triassic age. Sexten conglomerates and Tarvis breccia form the basis of the sequence which rests by an unconformity on crystalline basement and Lower Permian volcanic rocks (Bozen quartzporphyry). The conglomerates and breccia are overlain by Middle and Upper Permian Gröden Sandstone Formation and Bellerophon Formation, Lower Triassic Werfen Formation and some Lower Mid-Triassic formations. The entire depositional sequence shows onlapping towards the west. The basal sequence is subdivided in three tectonically controlled sedimentary cycles with members of sandstone, gypsum, dolomite and limestone. However, a significant regression in the upper part of the Bellerophon Formation could be attributed to a eustatic sea-level change [24].

The third-order cyclicity continues into the Werfen Formation, creating a sequence of stratigraphic members with an articulate alternation of

carbonates and terrigeneous deposits. Basic alterations occur in the lithofacies, such as a striking decrease of biogenes and bioturbation and a significant increase of tempestites towards the hanging wall. On top of this, cycles along the Permian–Triassic boundary indicate a sharp increase of amplitudes to approximately 10 m for the sea-level changes. A current event in the entire Dolomite region caused an out-wash of the fossiliferous Fusulina packstones and sedimentation of oolithic grainstones, which form the base of the Tesero horizon. This wedge-shaped unit is about 25 m thick in the Trento area, but thins gradually eastward and northward [2]. The Tesero Oolith was deposited during a transgression, comprising the switch between the Bellerophon Formation and the Mazzin member of the Werfen Formation.

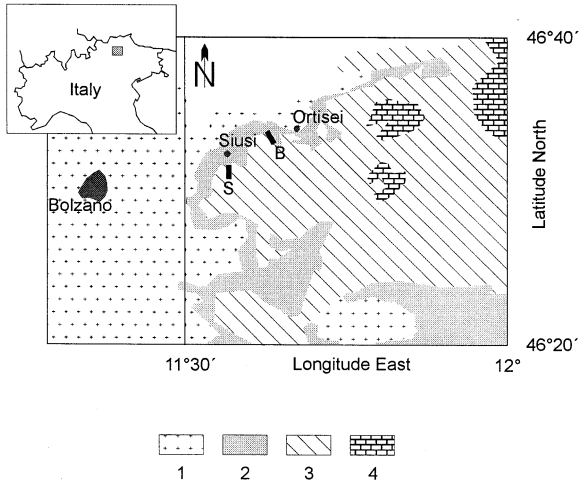


Fig. 1. Simplified geological map of the study area showing the locations of the palaeomagnetic sampling sites Siusi (S) and Bulla (B). 1, Permian granitoids; 2, clastic and shallow marine deposits (Middle Permian to lowermost Anisian); 3, Triassic basinal and platform deposits; 4, Jurassic platform deposits.

3. Sampling sites

The sampling sites are located approximately 20 km east of Bolzano (Bozen) in the Southern Tyrol in Italy (Fig. 1). The Siusi section is located in the Gola di Siusi (Seiser Klamm) along the Rio Bianco south of Siusi (46.517°N, 11.568°E). The Bulla section outcrops alongside the road which joins Ortisei to Bulla at an altitude of approximately 1500 m (46.558°N, 11.628°E). Both sections expose profiles from the uppermost part of the Permian Bellerophon Formation across the Permian–Triassic boundary unto the lower part of the Campil member of the Lower Triassic Werfen Formation. Black fossiliferous limestones are dominant in the upper part of the Permian Bellerophon Formation, with a topmost segment consisting of dark marls and limestones with fusulinids and other foraminifers, calcareous algae and brachiopods. The Bellerophon Formation is overlain by oolithic units of the Tesero Horizon of the Triassic Werfen Formation with no evidence for a stratigraphic gap. Most of the Werfen Formation was deposited in a shallow shelf-setting that was generally below fair-weather wave-base but was frequently affected by storm waves. The Mazzin, Seis and Campil members of the Werfen Formation are mainly shallow marine deposits, characterised by oolithic calcarenites, arenaceous limestones and sandstones. Centimetre to decimetre thick storm layers occur with a graded bedding, wave ripples on top of the beds and even and hummocky lamination. This deposits are interrupted by strata that indicate deposition on supratidal mud-flats. Such marls, marly limestones and fine-grained siltstones make up the Andraz Horizon and the uppermost parts of Seis and Campil members.

4. Methods

Oriented cores of 2.5 cm diameter were collected using a portable coring apparatus with a non-magnetic hollow drill bit. Standard 2.2 cm long specimen were cut from each core for palaeomagnetic laboratory analyses. Isothermal remanent magnetisation (IRM) acquisition and backfield experiments together with thermal demagnetisation of IRM aided identification of the magnetic mineral content of a representative number of pilot specimens. Specimens were subjected to detailed stepwise demagnetisation procedure (alternating field and/or thermal treatment). During thermal demagnetisation, the bulk susceptibility

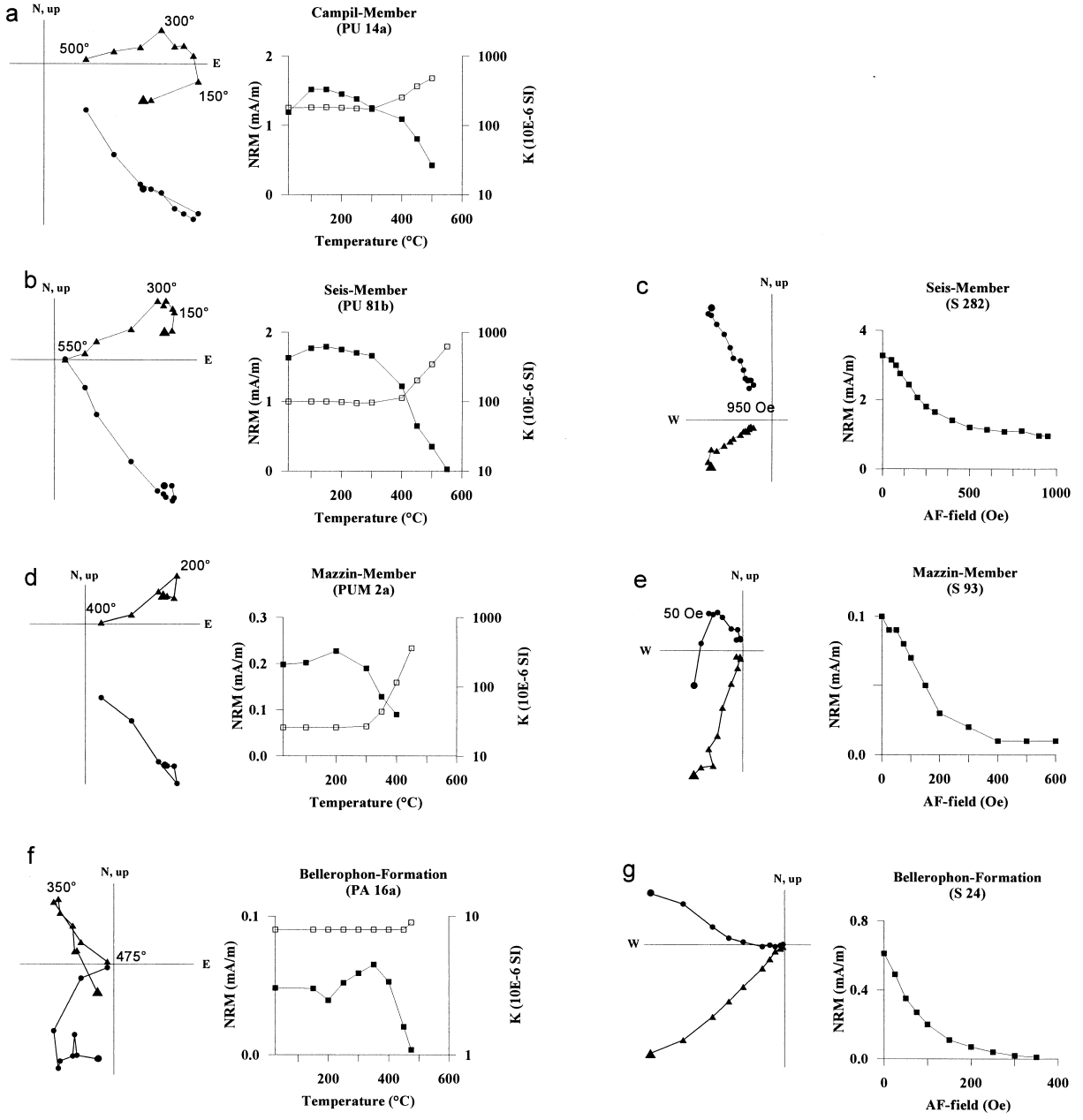


Fig. 2. Representative Zijderfeld diagrams showing the NRM vector during thermal demagnetisation (left column) and alternating-field demagnetisation (right column). Circles and triangles represent projection on the horizontal and vertical planes (NRM plotted with larger symbols). Change of remanent magnetisation (filled squares) and susceptibility (open squares) with temperature is shown.

of the specimens was routinely measured to observe possible mineral transformations [25]. Palaeomagnetic data analyses included principal component analysis [26] based on visual inspection of orthogonal projections [27]. Accompanying texture measurements by means of anisotropy of magnetic susceptibility [28] aimed at determining the origin of the magnetic fabric with respect to the sedimentary bedding planes. All measurements were carried out in the Palaeomagnetic Laboratory of the University of Leoben. Natural remanent magnetisation was measured on a three-axis cryogenic magnetometer with in-line degausser (2G Enterprises) and a Geofyzika JR-4 Spinner Magnetometer. Geofyzika KLY-2 and Digico instruments were used for measuring low-field magnetic susceptibility and its anisotropy.

5. Laboratory measurements and results

5.1. Demagnetisation behaviour and remanence carriers

Both the intensity and stability of the natural remanent magnetisation (NRM) of the samples were found to vary strongly in accordance with the different rock types. Moreover, there was considerable variability as to coercivity and unblocking spectra. Intensities of NRM and susceptibilities were generally low in the Bellerophon Formation and the lower Werfen Formation and higher in the upper part of the sections, indicating a higher concentration of magnetic minerals in the Campil and Seis members. NRM intensities varied in the range between 0.01 and 19 mA/m. The average low-field volume susceptibility was 56 (-1 to 188) $\times 10^{-6}$ SI and 44 (-3 to 264) $\times 10^{-6}$ SI for specimens from the Bulla and the Siusi sections, respectively. During thermal treatment, specimens from the upper parts of the sections typically showed mineral transformations. Susceptibilities increased in orders of magnitude during thermal demagnetisation at temperatures above 500°C. Therefore, in many cases the thermal demagnetisation of NRM could not be continued for temperatures exceeding 550°C. The demagnetisation data from the Bulla section gave evidence for the presence of high-coercivity components, which could be attributed to goethite, indicating a higher degree of weathering in the samples from the Bulla section compared with those of the Siusi section. Accordingly, only thermal treatment was applied to specimens from the Bulla section, while both alternating-field and thermal treatment of specimens from the Siusi section yielded good results.

Well-defined demagnetisation paths with up to three components of NRM and a good separation of the unblocking temperatures could be obtained from specimens from the Campil member. The first component was thermally demagnetised at 100°C, whereas the second and third component unblocked at 400 and 550°C, respectively (Fig. 2a). Different compositions of the magnetic mineral associations were reflected in various kinds of responses to the acquisition of isothermal remanent magnetisation (IRM). Some specimens rapidly gained remanence at low magnetising fields and entirely saturated at less than 0.7 T, others acquired half of their saturation IRM at only 0.1 T, but slowly continued to gain remanence at higher fields, without reaching full saturation even at the highest field available (Fig. 3). This behaviour suggested that goethite carried the low temperature component, whereas the high temperature components resided in two different phases of magnetite. The third component, which typically decayed towards the origin, could be regarded as primary characteristic remanence direction vectors with southeasterly directions and shallow, mostly negative inclinations after tilt correction, indicating a reversed polarity zone of the Earth-magnetic field.

Although there was variability as to the relative contributions of the respective remanence components, the same magnetic mineral associations were present in specimens from the Seis member. Accordingly, the same kinds of responses to exposure to magnetising fields were observed during IRM acquisition (Fig. 3). As a general rule, higher values of remanence intensities were typically found in specimens from the upper part of the Seis member and the influence of a low-temperature and high-coercivity component, suggesting goethite, increased towards the footwall. How-

ever, a high-temperature and low-coercivity component, suggesting magnetite, was obtained in all specimens. Again, orthogonal projections showed steep northerly components, indicative of a present-day-field overprint, which could be removed during thermal demagnetisation at temperatures below 400°C (Fig. 2b) or with alternating field strengths of 5 mT (Fig. 2c). The remaining component, characterised by an unblocking temperature of 550°C, yielded two antipodal groups of directions: southeasterly magnetisations with shallow negative (and some positive) inclinations occurred both in the uppermost part and the lower part, whereas northwesterly directions were typically found in the middle part, suggesting a normal polarity zone within the Seis member.

The quality of the demagnetisation data from the underlying strata differed substantially from those presented above. Specimens from the Mazzin member unto the Bellerophon Formation were characterised by generally lower susceptibilities and lower intensities of NRM. Most of the specimens could not be completely demagnetised due to mineral transformations, which occurred at higher temperatures, and very low NRM intensities. During IRM acquisition of specimens from the Mazzin member, about 90% of the saturation remanence was typically acquired in an ambient field of 0.3 T (Fig. 3). Specimens from the upper part of the Mazzin member contained two different directions of NRM, one of which was a present-day field overprint (in situ mean direction: declination = 5°, inclination = 65°), which was removed at 200°C. Subsequent thermal demagnetisation steps up to 400°C yielded a second magnetisation vector with a southeasterly direction and negative inclination after tilt correction, indicating an reversed polarity zone. Since the decay was not pointing towards the origin (Fig. 2d), the vectors had to be analysed by calculating the difference vectors between 200 and 400°C. Specimens from the lower part of the Mazzin member did not show a good separation of the blocking temperatures. Alternating-field demagnetisation of the specimens yielded directions with southwesterly declination and a steep positive inclination (Fig. 2e). Subsequently, a well defined final component (northwest with positive inclination) could be observed in both sections by means of different demagnetisation methods, indicating another normal polarity zone in the hanging wall of the Tesero Horizon.

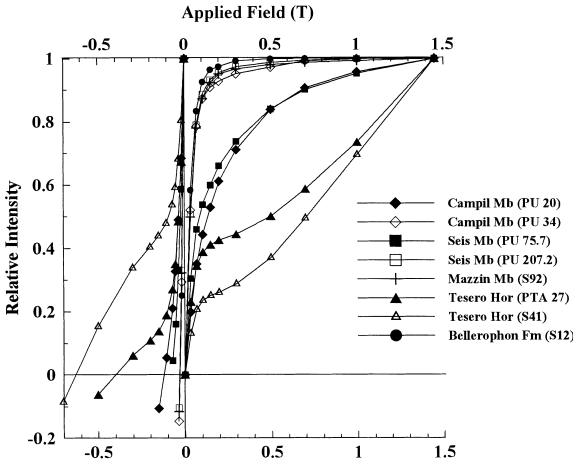


Fig. 3. Isothermal remanent magnetisation (IRM) and back-field-acquisition. The majority of the samples were characterised by a low-coercivity phase with varying contributions from a high-coercivity phase, most likely goethite. High coercivity and Curie temperatures above 600°C indicated the presence of hematite in specimens from the Tesero Horizon.

A significant change in magnetic mineral composition was observed during IRM acquisition and thermal demagnetisation of saturation remanence of specimens from the Tesero Horizon. Specimens rapidly acquired remanence at low magnetising field strengths, but continued to gain remanence at higher fields. None of the specimens was entirely saturated at 1.45 T (Fig. 3). During subsequent thermal demagnetisation of saturation remanence, a magnetic phase with an unblocking temperature higher than 600°C was observed, suggesting that hematite contributed considerably to the high-coercivity spectra. The NRM of most specimens carried pervasive present-day field remagnetisations (in situ mean direction: declination = 350°, inclination = 63°).

Specimens from the uppermost Permian Bellerophon Formation showed random components at temperatures up to 350°C, while the final component was defined by a demagnetisation path towards the origin at higher temperatures (Fig. 2f) and alternating field strengths (Fig. 2g). Speci-

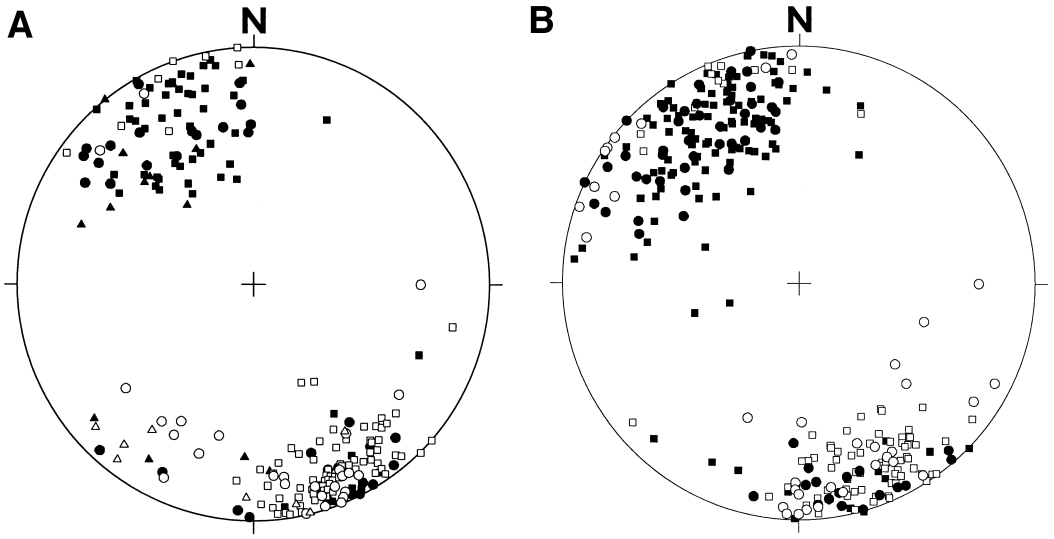


Fig. 4. Stereographic projection of the characteristic remanence magnetisation (ChRM) directions after tilt correction for the Bulla (A) and Siusi (B) sections. Equal area projection with full symbols for lower hemisphere and open symbols for upper hemisphere. Squares, circles and triangles represent Class 1, 2 and 3 results, respectively.

mens from a few beds immediately below the Tesero Horizon showed magnetisation vectors with southerly declinations and negative inclinations, suggesting a zone of reversed polarity below the Permian–Triassic boundary in both sections. The samples from the underlying units, which originated from the Siusi section only, showed again normal polarity.

5.2. Characteristic remanence directions

The directions of the primary components, which were used for the magnetostratigraphic zonation, are presented in a stereographic projection in Fig. 4. Characteristic remanent magnetisation directions (ChRM) for single samples were determined by principle component analyses of the magnetisation components observed during thermal or alternating-field demagnetisation. The quality of the demagnetisation data varied strongly in accordance with the different rock types. Taking this into account, a qualification of the ChRM data with respect to the demagnetisation behaviour has been applied. Results from 141 specimens from the Bulla section that yielded consistent demagnetisation paths towards the origin with an unblocking temperature higher than 500°C, indicating magnetite as the primary carrier of the remanence, were assigned to Class 1. The second group was also characterised by demagnetisation paths towards the origin. It comprised 60 results obtained in the temperature range up to 500°, in particular from specimens which could not be demagnetised further due to mineral transformations at higher temperatures. Class 3 represents 22 results from well defined difference vectors, where demagnetisation paths did not show decay towards the origin. Unfortunately, 148 samples from the Bulla section had to be rejected, because they merely carried a present-day field overprint or yielded uninterpretable demagnetisation patterns. Directions that were assigned to Class 3 results were characterised by a considerable scatter, while the grouping improved significantly for the Class 2 and Class 1 results. The ChRM data from the Siusi section were treated accordingly: Class 1 represented results from thermal or alternating-field demagnetisation of 181 specimens, that indicated magnetite as the primary carrier of the remanence. Class 2 comprised all remaining results obtained from successful thermal demagnetisation of 96 specimens, where it was not possible to attribute particular magnetisation components to particular magnetic

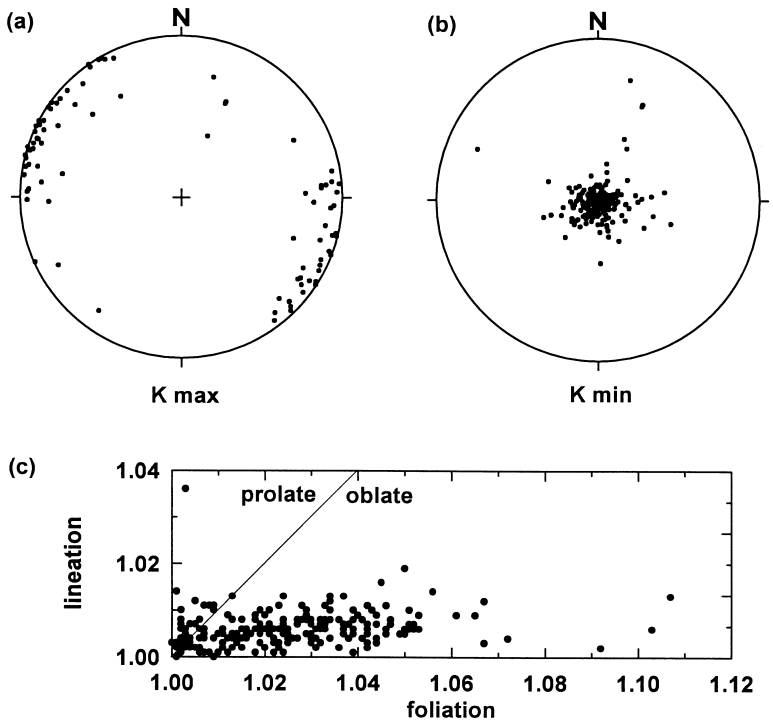


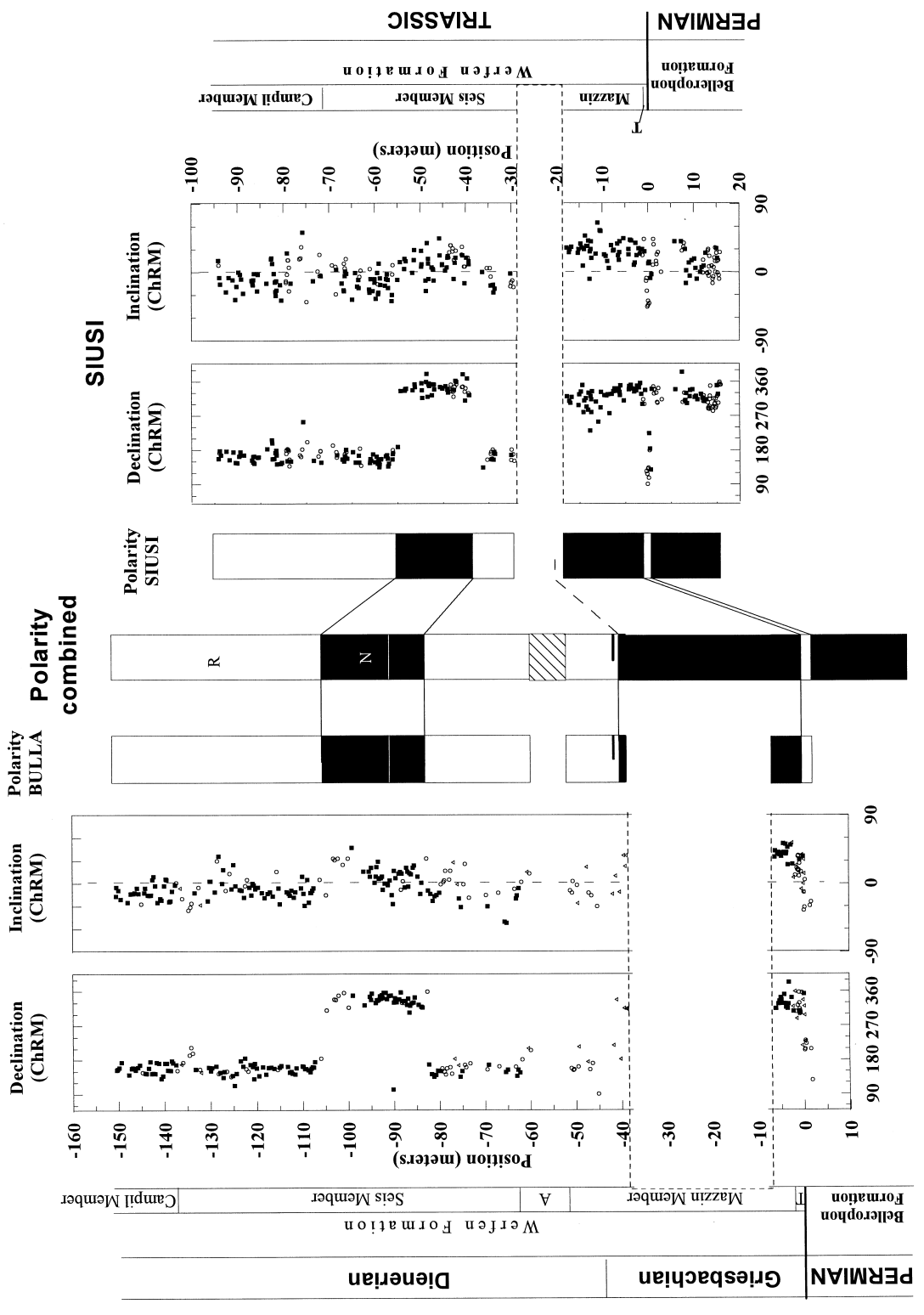
Fig. 5. Stereographic projection of the maximum (a) and minimum (b) susceptibility axes after tilt correction, and magnetic lineation versus foliation (c) for the Bulla section. Equal area projection, lower hemisphere. The preferred orientation of the maximum susceptibility directions indicated alignment by a weak or moderate current.

minerals. Unfortunately, more than 50% of the samples yielded uninterpretable demagnetisation patterns. Again, grouping improved significantly for the Class 1 results.

Only Class 1 results were included in the calculation of mean characteristic magnetisation directions [29] for the Bulla and Siusi sections. The normal polarity mean direction, regardless of the actual polarity, has a declination of 335.4° and an inclination of 15.5° after tilt correction for the Bulla section and a declination of 335.3° with an inclination of 18.6° for the Siusi section. There is a significant trend in the palaeomagnetic directions between the normal and reverse polarity data, with shallower inclinations for the reverse polarity mean directions (Table 1). In both cases, the reversal test of McFadden and McElhinny [30] is negative ($\gamma = 9.2° > \gamma_C = 6.6°$ for Bulla and $\gamma = 11.6° > \gamma_C = 6.4°$ for Siusi). The deflection from ideal antipodality suggests that a component with a steep downward inclination was not completely isolated from the primary magnetisation components. The bedding attitudes of the strata are very similar in Bulla and all horizontal in Siusi. Thus, fold tests within each site are not significant. However, the grouping of the combined mean direction for Bulla and Siusi improves to a great extent after tilt correction, indicating that the magnetisation was acquired prior to the tilting of the strata in Bulla. (Table 1).

→

Fig. 6. Palaeomagnetic results and magnetic polarity zonation for the Bulla and Siusi sections. Reversed (white; R) and normal (black; N) polarity obtained from the distribution of characteristic remanence magnetisation (ChRM) directions. A, Andraz Horizon; T, Tesero Horizon. Filled squares, open circles and open triangles represent Class 1, 2 and 3 results, respectively. Position (0) at the synchronous current event in both sections.


PERMIAN
Griesbachian
Dienerian
TRIASSIC
Werfen Formation
Bellerophon Formation
Mazzin Member
T
A
Seis Member
Campil Member
Position (meters)
Declination (ChRM)
Inclination (ChRM)
Polarity BULLA
Polarity combined
Polarity SIUSI
SIUSI
R
N

5.3. Anisotropy of magnetic susceptibility

Anisotropy of low-field magnetic susceptibility gave reliable evidence for a primary sedimentary origin of the magnetic fabric. Most of the specimens did not show a statistically significant anisotropy at all and the remaining samples (all from the Bulla section) yielded susceptibility ellipsoids, that indicated depositional fabrics (Fig. 5): the maximum susceptibility axes (K_{max}) were aligned within the macroscopic bedding planes, while the minimum susceptibility axes (K_{min}) formed a cluster around the poles to the bedding planes. The mean direction for the minimum susceptibility had an inclination of 88° (187 significantly anisotropic K_{min} axes, $\alpha_{95} = 1.6°$) after tilt correction, indicating a magnetic foliation plane, which was, within the statistical limits, exactly parallel to the bedding planes. The distribution of maximum susceptibility directions showed a preferred orientation along an east–southeasterly axis (84 significantly anisotropic K_{max} axes, declination = 112°, inclination = 0°, $\alpha_{95} = 4.7°$). Susceptibility ellipsoids were spherical or weakly oblate, except for some samples with weakly prolate fabrics. Magnetic lineation (K_{max}/K_{int}) was typically below 1.02 with an average lineation of 1.006 and the average magnetic foliation (K_{int}/K_{min}) was 1.023. The results suggested the presence of a weak or moderate palaeocurrent at the time of deposition. Most likely, the palaeocurrent produced the weak lineation by aligning the long axes of prolate grains parallel to the direction of flow.

6. Discussion

The mean palaeomagnetic direction (Table 1) for the Bulla and Siusi sections (declination (D) = 335.3°, inclination (I) = 17.1°, $\alpha_{95} = 6.8°$) is consistent with results from earlier palaeomagnetic studies of Permian and Triassic sediments and volcanites in the Southern Alps [31–35]. These results imply very little net rotation of the Dolomites relative to Africa since Permian time [35], as well as a motion towards higher latitudes between the Lower Permian ($D = 320°$, $I = 14°$, $\alpha_{95} = 8°$ [33]) through Early Triassic ($D = 331°$, $I = 13°$, $\alpha_{95} = 7°$ [35] and the Middle and Late Triassic ($D = 329°$, $I = 25°$, $\alpha_{95} = 9°$ [32]), indicating deposition and remanence acquisition at a shallow northerly palaeolatitude.

An earlier magnetostratigraphic study of Lower Triassic sections within the Werfen Formation in the Dolomites yielded a magnetic polarity succession that was described as being consistent with results from the Canadian Arctic [36]. The obtained results from the Bulla and Siusi sections established well defined groups of normal and reversed primary magnetisations, enabling a magnetic zonation of the sediments (Fig. 6). Importantly, the current event (i.e. boundary event) at the base of the Tesero Horizon can be regarded as synchronous boundary in a strict sense. In contrast, chronostratigraphic positions of the lithostratigraphic boundaries between members of the Werfen Formation may vary between sites, since Tesero oolith and Mazzin member interfinger. Ac-

Table 1
Palaeomagnetic data from this study

	D_g	I_g	α_{95}	k	D_s	I_s	α_{95}	k	n	Plat	Plon
Bulla normal	335.1	−1.0	5.2	15.1	335.6	21.0	5.7	12.6	54		
Siusi normal	331.0	21.7	4.7	9.8	331.0	21.7	4.7	9.8	103		
Bulla reverse	154.9	7.3	3.5	19.5	154.7	−11.5	3.5	19.3	87		
Siusi reverse	160.6	−14.6	4.3	15.0	160.6	−14.6	4.3	15.0	78		
Bulla	335.0	−5.0	2.9	17.3	335.4	15.5	3.8	11.0	141	46.0	228.0
Siusi	335.3	18.6	3.3	11.0	335.3	18.6	3.3	11.0	181	47.5	229.1
Bulla+Siusi	335.1	6.8	53.9	23.7	335.3	17.1	6.8	1365.3	(2)	46.7	228.6

D_g, I_g mean declination and inclination (in situ), α_{95} confidence interval at the 95% level, k precision parameter, D_s, I_s mean declination and inclination (tilt corrected), n number of samples (sites), Plat, Plon latitude (°N) and longitude (°E) of the palaeopole.

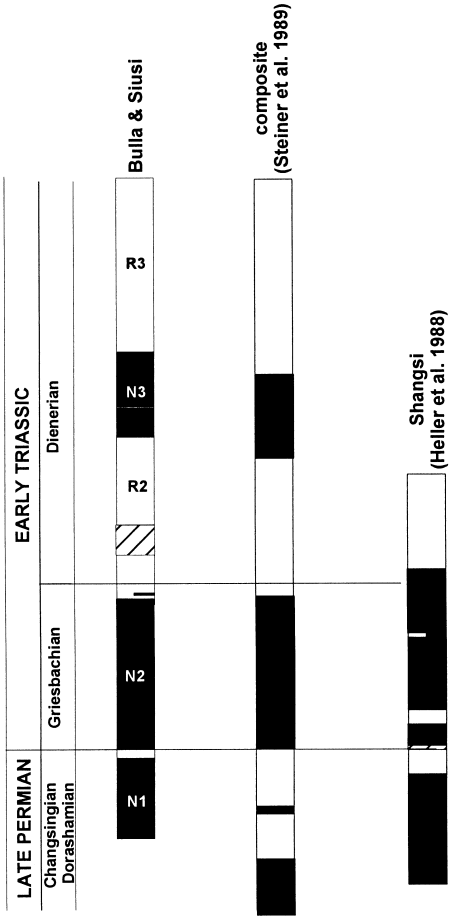


Fig. 7. Comparison of Permian–Triassic magnetic polarity zonation of sections from Italy (Bulla and Siusi, this study) with the composite section of Steiner et al. [1] and results from South China. [14]. Reversed (white) and normal (black) polarity.

cording to the palaeogeographic position within the Triassic Werfen sea, sedimentation rates were generally higher in the Bulla section than in the Siusi section (see Figs. 1 and 6).

Most of the upper part of the Permian Bellerophon Formation, sampled in the Siusi section, carried normal polarity magnetisations, while a zone of reversed polarity occurred in both sections below the boundary with the Tesero Horizon. The next polarity change from reverse to normal polarity occurred shortly above this boundary, within the Tesero Horizon in the Bulla and Siusi sections. All samples from the Mazzin member in the Siusi section together with samples from the lower part of the Mazzin member sampled in the Bulla section showed normal polarity magnetisation. In the uppermost part of the Mazzin member, mainly reversed polarity occurred again, except in some places, where zones of normal polarity are represented by Class 3 results. A reversed polarity zone was observed throughout the lower part of the Seis member. The samples from the middle part of the Seis member were characterised by normal polarity magnetisation, except for a thin zone of reversed polarity, represented by a single Class 1 result in the Bulla section only. Finally, the samples from the upper part of the Seis member together with the samples from the Campil member on top of the sequences carried reversed polarity magnetisation, except for a single sample from the Siusi section at position −75.7 m with an atypical direction ($D = 254°$, $I = 52°$), that could be an indication for a small excursion.

The combined magnetostratigraphic results for the Bulla and Siusi sections result in a magnetic polarity zonation (Fig. 7) which is in good accordance with the composite Late-Permian/Early-Triassic magnetic polarity time scale of Steiner et al. [1], but discrepancies with other results have to be mentioned. The two polarity zones observed in the Bellerophon Formation (N1, R1) correspond to the topmost part (16 m in the Siusi section) of a Late Permian sequence with a total thickness of up to 350 m in the study area. The reversed polarity zone (R1) at the top of the Permian sequence was observed also in many other regions [1,12–18]. The basal normal polarity zone in the Triassic (N2) corresponds to most of Griesbachian time in agreement with [1], whereas other authors reported additional chrons within the Griesbachian [14,15,18]. Furthermore, a reversed polarity zone (R2) across the Griesbachian–Dienerian boundary, as well as the normal polarity zone (N3) in the lower part of the Dienerian and the subsequent reverse polarity zone (R3) in the upper part of the Dienerian are in accordance with observations from South China [1] and the Canadian Arctic [18].

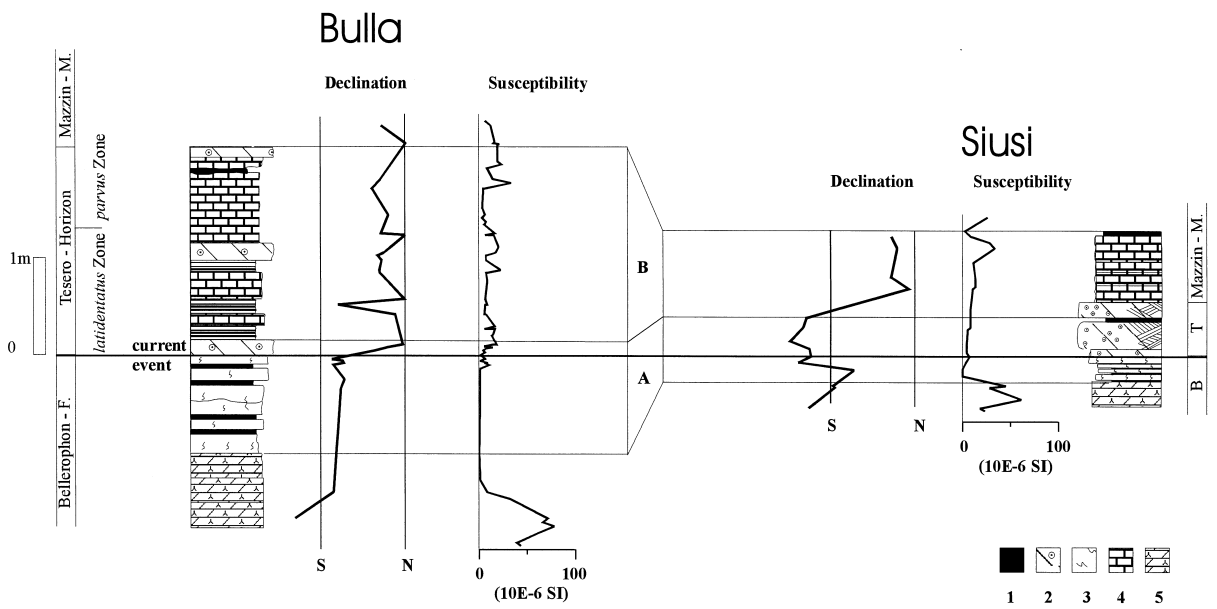


Fig. 8. Position of the polarity change from reverse (southerly declination; S) to normal (northerly declination; N) polarity at the Permian–Triassic boundary in the Bulla and Siusi sections. A and B, tectonically controlled sedimentary cycles. 1, black carbonaceous claystone and green silty marlstone; 2, oolithic grainstone; 3, fossiliferous bioturbated packstone and wackestone; 4, mudstone; 5, dolomite with root traces. The current event can be regarded as synchronous. Conodont biozonation after Farabegoli and Perri [37]. The onset of sedimentary cycle B marks the beginning of the basal Triassic normal polarity zone. A drop in susceptibility occurs in the Bellorophon Formation below the boundary.

7. Permian–Triassic boundary

The exact chronostratigraphic position of the Permian–Triassic boundary in the Southern Alps and the type of contact between uppermost Bellerophon Formation and lowermost Werfen Formation is still under discussion [2]. A significant drop in taxonomic diversity at the Bellerophon Formation–Werfen Formation contact corresponds to the current event in the entire Dolomite region, which caused an out-wash of the Permian fossiliferous Fusulina packstones and sedimentation of oolithic grainstones, that form the base of the Tesero Horizon. Some authors [37] believe that the formational contact is unconformable and that a gap representing early Dorashamian through earliest Griesbachian time occurs between the two formations, whereas several other authors [2–5] emphasise a gradual transition in lithology and palaeontology with no time gap at the boundary. This interpretation is strongly supported by the recent discovery of *Paratirolites*, that corresponds to late Dorashamian age, in the Bellerophon Formation near Santa Cristina Valgardena [38].

Broglio Loriga and Cassinis [2] summarised the criteria that should be used in defining the Permian–Triassic boundary in the Southern Alps: below the lithic boundary, several Permian brachiopods (*Comelicania, Janiceps*) disappear. Flourishing of fungal remains (*Tympanocysta*) is followed by a drop in taxonomic diversity at the Bellerophon Formation–Werfen Formation contact. The lower part of the Tesero Horizon is characterised by mixed fauna with a co-occurrence of the highest Permian brachiopods and Werfen-like fossils. It has been referred to the *latidentatus* conodont biozone owing to the co-occurrence of *Hindeodus latidentatus*, *Isarcicella ? prisca* and other elements. Subsequently fungal remains and certain brachiopods (*Orthotetina, Ombonia*) disappear and Triassic acritarchs develop 1–2 m above the lithostratigraphic boundary. The peak of the Permian mass extinction coin-

cides rather well with the lithostratigraphic boundary [2]. This interpretation is supported by a recent investigation, in which parts of the Bulla section were resampled for conodont biostratigraphy. *Hindeodus parvus* first occurs 1.30 m above the lithostratigraphic boundary. The appearance of this species defines the lower limit of the *parvus* zone that is the first Scythian conodont biozone [39].

In accordance with other magnetostratigraphic studies across the Permian–Triassic boundary in many regions of the world [1,12–18], we observed a zone of reversed polarity that corresponds to latest Dorashamian time at the top of the Permian sequence. The polarity change from reverse to normal polarity occurs shortly above the base of the Werfen Formation (Fig. 8). The onset of the subsequent sedimentary cycle marks the beginning of the basal Triassic normal polarity zone in the Bulla and Siusi sections. The record of geomagnetic polarity across the Permian–Triassic boundary confirmed the apparent continuity as found by sedimentologic investigations which emphasise a gradual lithic transition with no evidence for a time gap at the boundary.

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